Matching and Propensity Scores

ECON726

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Housekeeping

Today

- Discuss paper selection
- Finish linear regression
- Introduction to matching
- Review examples in jupyter notebooks (regression; matching)

Homework

- Read MHE 5.1. and 5.2. (Fixed effects and differences-in-differences)
  - We will focus on diff-in-diff next class, but many of your papers use fixed effects
- Skim Causal Inference Mixtape:
  - Chapter 8 : Panel designs (`https://mixtape.scunning.com/09-difference_in_differences`
  - Chapter 9 : Difference-in-differences fundamentals (`https://mixtape.scunning.com/09-difference_in_differences`)

Empirical project

Paper selection: Due **September 28** [5% of final grade]

- Done!

Project plan: Due **October 13** [15% of final grade]

- Done!

Replication results: Due **November 2**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 1** & **December 8** [10% of final grade]

- Prepare a 10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 11** [40% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Empirical project

General feedback:

- **These were generously graded**. I wanted to give people credit for staying on track. Standards are higher on the final paper!
- **Read the comments on your assignment!** Note: I use negative grading. I'm sorry if the comments seems harsh!

Overall feedback:

- For your final paper, you need to motivate why your method is causal. The authors will say "I use differences-in-differences to identify..." You cannot get away with that. You need to motivate why yor method is causal. If you don't discuss the conditional independence assumption, you will lose a lot of points. **This will be graded harshly on your final project**
  - Part of that will include discussing what the ideal randomized experiment is.
  - You need to explain why this method is causal
  - Does SUTVA apply?
  - Limitations of data should be meaningful. Who aren't you looking at?
- Do **NOT** reference methodologies that you cannot explain why they are important.
  - These are long, published papers. If you cannot explain in plain language the robustness check, don't include it.
- You're expected to clearly understand everything you state in this paper
  - Imagine you are explaining to a coworker who has no idea about the context of your paper.
- You're expected to incorporate parts of the reading into your discussion of causality (and, likely, do readings from secondary sources)
- Include your paper title, authors etc. Ideally, also include the link! And ideally, in an appendix, include the original figures you're referencing!

Next assignments: replication and presentation

You are expected to replicate key results.

- If you simply upload the replication file for the paper, you will fail
- You need to distill these files to only **key results**
- I should effectively do **zero** data cleaning when I re-run your files. You should have a clean file, a few lines of code; that's it

When replicating your results, keep in mind your presentation.

- Your presentation can only be about 3-4 slides.
  1. 1 slide with intro/context/question
  2. 1-2 slide on key points of data/methodology
  3. 1-2 slide with key results
  4. 0-1 slide with extension

How do you distill this paper to key takeaways?

Outline

Matching

Matching estimators

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Matching

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Control for covariates using the Propensity Score

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Motivating example

Consider the universe of applicants to the U.S. military:

- Y_i: average earnings
- D_i: dummy variable indicating whether i served in the military

We may be interested in the average effect of treatment on the treated (ToT):

$$\mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]$$

Difference between:

- Average earnings of soldiers (observable)
- Average earnings of soldiers, had they not served (unobservable)

Naïve comparison OR **average predictive effect** OR **simple difference in outcomes**:

$$\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0]$$

- This is *observable* in the data
- Will be biased:

$$\mathbf{E}[Y_i|D_i = 1] - \mathbf{E}[Y_i|D_i = 0] = \underbrace{\mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y_i(0)|D_i = 1] - \mathbf{E}[Y_i(0)|D_i = 0]}_{\text{Selection bias}}$$

Motivating example: CIA

Conditional independence assumption:

$$Y_i(0), Y_i(1) \perp D_i | X_i$$

Assuming CIA, selection bias disappears after conditioning on X_i:

$$\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0) | D_i = 1]$$

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\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_i(0) | X_i, D_i = 0] = \mathbf{E}[Y_i(0) | X_i, D_i = 1]$$

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\end{aligned}$$

By CIA:

$$\mathbf{E}[Y_i(0) | X_i, D_i = 0] = \mathbf{E}[Y_i(0) | X_i, D_i = 1]$$

By definition of $Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$:

$$\begin{aligned}
\mathbf{E}[Y_i | X_i, D_i = 0] &= \mathbf{E}[Y_i(0) + D_i(Y_i(1) - Y_i(0)) | X_i, D_i = 0] \\
&= \mathbf{E}[Y_i(0) | X_i, D_i = 0] \\
\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_i(1) | X_i, D_i = 1]
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\mathbf{E}[Y_i | X_i, D_i = 1] &= \mathbf{E}[Y_i(1) | X_i, D_i = 1]
\end{aligned}$$

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1]}_{} - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1 \right]$$

Consider the parts of this estimator:

$\mathbf{E}[Y_i|X_i, D_i = 1]$: **For a given** X_i, take the average over all observed **treated** Y_i

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$\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

$$\delta_{ToT} = \mathbf{E}\left[\underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}\,\middle|\, D_i = 1\right]$$

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$\mathbf{E}[Y_i|X_i, D_i = 0]$: **For a given** X_i, take the average over all observed **untreated** Y_i

Define δ_X to be the average difference in observed Y_i for treated and untreated with a given value X_i:

$$\delta_X \equiv \underbrace{\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]}$$

- Example: difference in mean earnings by veteran status at each value of X_i
- Requires both treated and untreated units with $X_i = x$: this is called **common support**

$$\delta_{ToT} = \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] \,\middle|\, D_i = 1\right]$$
$$= \mathbf{E}\left[\delta_X \,\middle|\, D_i = 1\right]$$

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- Example: difference in mean earnings by veteran status at each value of X_i
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Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Let's say there was some variable X_i that represents an indicator of whether an applicant is "high ability" or "lower ability"

- $X_i = 0$ means i is low ability
- $X_i = 1$ means i is high ability

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$ Difference in observed Y_i for treated and untreated with $X_i = 0$
(Average difference in wages by veteran status for **low** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

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(Average difference in wages by veteran status for **low** ability i)

$\delta_{X_i=1}$ Difference in observed Y_i for treated and untreated with $X_i = 1$
(Average difference in wages by veteran status for **high** ability i)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

How do these elements connect to data?

$\delta_{X_i=0}$	Difference in observed Y_i for treated and untreated with $X_i = 0$ (Average difference in wages by veteran status for **low** ability i)
$\delta_{X_i=1}$	Difference in observed Y_i for treated and untreated with $X_i = 1$ (Average difference in wages by veteran status for **high** ability i)
$\mathbf{P}(X_i = 0\|D_i = 1)$	Probability $X_i = 0$ conditional on being treated (Fraction of veterans that are **low** ability)

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \delta_{X_i=0}\mathbf{P}(X_i = 0|D_i = 1) + \delta_{X_i=1}\mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

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Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect
- This is called **subclassification**: each value of X_i defines a **cell** (or stratum) (Mixtape)

Effect of treatment on the treated under CIA:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}\left[\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]|\, D_i = 1\right] \\ &= \mathbf{E}\left[\delta_X|\, D_i = 1\right]\end{aligned}$$

where

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Now instead of X_i being a dummy, assume X_i is the result of a test of ability/intelligence

- $X_i \in \{0, 1, \ldots, 100\}$ and measures ability

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x=0}^{100} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1) + \cdots + \delta_{100} \mathbf{P}(X_i = 100|D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- X_i-specific treatment-control comparisons, weighted together to produce a single overall treatment effect

Now suppose X_i isn't a single variable, but instead a vector of (discrete) variables

- Year of birth, test score, year of application to the military, educational attainment at the time of application

Think of each realized $X_i = x$ as a particular combination of all these variables

- For example, $X_i = z$ might describe an applicant born in 1970, who scored a 95 on an ability test, and graduated from college

We'll assume CIA: $\{Y_i(0), Y_i(1)\} \perp D_i | X_i$

- Intuitively, what does this mean? Is this a good assumption? Is it realistic?

The matching estimand can be written as:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[\delta_X | D_i = 1] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x | D_i = 1)\end{aligned}$$

Intuitively, what is this matching estimate?

- The δ_x are x-specific treatment-control comparisons
  - For $X_i = z$, compare average wage difference of veterans and non-veterans who were born in 1970, who scored a 95 on an ability test, and graduated from college
  - Difference in average wages by veteran status for those with the same X_i
- Weighted together to produce a single overall treatment effect
  - Calculate weights based on the distribution of characteristics X_i among veterans

Unconditional Average Treatment Effect

Conditional treatment effect on treated:

$$\begin{aligned}\delta_{ToT} = \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] &= \mathbf{E}[\delta_X|D_i = 1] \\ &= \sum_{x\in\{0,1\}} \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ &= \delta_0 \mathbf{P}(X_i = 0|D_i = 1) + \delta_1 \mathbf{P}(X_i = 1|D_i = 1)\end{aligned}$$

- How much the typical soldier gained or lost as a result of enlisting

Can also write unconditional average treatment effect:

$$\begin{aligned}\delta_{ATE} &= \mathbf{E}\left[\mathbf{E}[Y_i(1)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0]\right] \\ &= \mathbf{E}[Y_i(1) - Y_i(0)] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x)\end{aligned}$$

- How much the typical applicant would gain or lose as a result of enlisting

Matching and regression

Matching and regression are related:

- Both are comparing outcomes (differences in Y_i) over treatment groups, controlling for covariates X_i
- These methods differ in implementation

Regression estimates of matching strategy:

$$Y_i = \sum_x d_{ix}\alpha_x + \delta_R D_i + e_i$$

where:

- $d_{ix} = \mathbf{1}[X_i = x]$: dummy variable that equals 1 if $X_i = x$, 0 otherwise
- α_x: regression effect for $X_i = x$
- δ_R: regression estimand

Matching and regression differences

In general, estimates from matching and from regression will often be similar, but not identical.

- Regression can be thought of as a type of matching strategy
  - Average effects are calculated by covariates X: δ_X
  - These effects are weighted to produce a weighted average
- Difference in how weights are determined. For ATE on treated:
  - Matching: weights are determined by distribution of covariates on treated
  - Regression: uses treatment-variance weighted average
  - These will differ unless treatment is independent of covariates

Note: neither regression nor covariate matching estimands give any weights to covariate cells that do not contain both treated and control observations

- This is the **common support** condition from before

Matching

Matching estimators

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Curse of dimensionality

Each additional covariate multiplies the number of cells:

- 2 binary covariates: $2^2 = 4$ cells
- 10 binary covariates: $2^{10} = 1{,}024$ cells
- Veterans: 10 birth years $\times$ 101 test scores $\times$ 4 education levels $= 4{,}040$ cells

Two problems:

1. Computational cost: more cells to compute and track
2. **Main problem:** many cells contain treated units but no untreated units
   - δ_x cannot be computed for those cells
   - This is a failure of **common support**
   - With a continuous covariate (e.g., income), exact matches are nearly impossible

(Mixtape)

Exact matching

Subclassification compares cells. **Exact matching** compares units:

- For each treated unit i, find an untreated unit $j(i)$ with the same covariates: $X_{j(i)} = X_i$
- Use $Y_{j(i)}$ to impute i's missing counterfactual $Y_i(0)$

$$\hat{\delta}_{ToT} = \frac{1}{N_T} \sum_{i:D_i=1} \left(Y_i - Y_{j(i)} \right)$$

- Notation: $j(i)$ is the untreated unit matched to treated unit i; $Y_{j(i)}$ is its outcome; N_T is the number of treated units
- If several untreated units match, average their outcomes
- This is a sample version of $\sum_x \delta_x \mathbf{P}(X_i = x | D_i = 1)$
- Still requires common support: every treated unit needs an exact match

(Mixtape)

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500			
2	29	12,250	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500			
2	29	12,250	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500			
2	29	12,250	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	**18**	**9,500**	11	20	8,500	**24**	**18**	**8,050**
2	29	12,250	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	**18**	**8,050**			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	**29**	**12,250**	12	27	10,075			
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	**16**	**29**	**10,525**			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	**29**	**12,250**	12	27	10,075	**16**	**29**	**10,525**
3	24	11,000	13	21	8,725			
4	27	11,750	14	39	12,775			
5	33	13,250	15	38	12,550			
6	22	10,500	**16**	**29**	**10,525**			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	33	13,250	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	**33**	**13,250**	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	18	33	11,425			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	**33**	**13,250**	15	38	12,550			
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	**18**	**33**	**11,425**			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	**33**	**11,425**			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	**33**	**13,250**	15	38	12,550	**18, 21**	**33**	**11,425**
6	22	10,500	16	29	10,525			
7	19	9,750	17	39	12,775			
8	20	10,000	**18**	**33**	**11,425**			
9	21	10,250	19	24	9,400			
10	30	12,500	20	30	10,750			
			21	**33**	**11,425**			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$
Tie: unit 5 has two exact matches, so average them: $(11{,}425 + 11{,}425)/2 = 11{,}425$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	33	13,250	15	38	12,550	18, 21	33	11,425
6	22	10,500	16	29	10,525	23	22	8,950
7	19	9,750	17	39	12,775	27	19	8,275
8	20	10,000	18	33	11,425	11	20	8,500
9	21	10,250	19	24	9,400	13	21	8,725
10	30	12,500	20	30	10,750	20, 28	30	9,875
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	11,075	Mean	31.95	11,101.25			

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.6–5.7)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**		
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings
1	18	9,500	11	20	8,500	24	18	8,050
2	29	12,250	12	27	10,075	16	29	10,525
3	24	11,000	13	21	8,725	19	24	9,400
4	27	11,750	14	39	12,775	12	27	10,075
5	33	13,250	15	38	12,550	18, 21	33	11,425
6	22	10,500	16	29	10,525	23	22	8,950
7	19	9,750	17	39	12,775	27	19	8,275
8	20	10,000	18	33	11,425	11	20	8,500
9	21	10,250	19	24	9,400	13	21	8,725
10	30	12,500	20	30	10,750	20, 28	30	9,875
			21	33	11,425			
			22	36	12,100			
			23	22	8,950			
			24	18	8,050			
			25	43	13,675			
			26	39	12,775			
			27	19	8,275			
			28	30	9,000			
			29	51	15,475			
			30	48	14,800			
Mean	24.3	**11,075**	Mean	31.95	11,101.25	Mean	24.3	**9,380**

Simple difference in outcomes: $11{,}075 - 11{,}101.25 = -26.25$

Exact matching: $\hat{\delta}_{ToT} = 11{,}075 - 9{,}380 = 1{,}695$

Nearest neighbor matching

When exact matches don't exist, use the **closest** untreated unit instead:

- $j(i)$ is now the untreated unit whose $X_{j(i)}$ is nearest to X_i
- Same estimator as exact matching:

$$\hat{\delta}_{ToT} = \frac{1}{N_T} \sum_{i:D_i=1} \left(Y_i - Y_{j(i)} \right)$$

Trade-off: we no longer need exact matches, but $X_{j(i)} \neq X_i$

- The gap between X_i and $X_{j(i)}$ is a **matching discrepancy**
- Larger discrepancies $\implies$ more bias in $\hat{\delta}_{ToT}$
- More covariates $\implies$ nearest neighbors are farther away (curse of dimensionality again)

(Mixtape)

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499				
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499				
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	**23**	**17,728**	11	49	15,499				
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	**23**	**15,238**				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	**23**	**17,728**	11	49	15,499	**30**	**23**	**15,238**	**0**
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	**23**	**15,238**				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566				
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	**26**	**17,770**	12	54	15,566	**20, 21**	**27, 25**	**15,265**	**1**
3	23	17,736	13	47	15,487				
4	27	17,777	14	49	15,501				
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	**20**	**27**	**15,281**				
			21	**25**	**15,249**				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

No one is age 26; average the two nearest (ages 25, 27): $(15{,}281 + 15{,}249)/2 = 15{,}265$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	31	17,832	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	**31**	**17,832**	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	**31**	**17,832**	15	55	15,554				
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	**32**	**15,321**				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	**31**	**17,832**	15	55	15,554	**22**	**32**	**15,321**	**1**
6	25	17,758	16	44	15,448				
7	24	17,744	17	34	15,352				
8	26	17,764	18	50	15,509				
9	20	17,706	19	45	15,469				
10	30	17,805	20	27	15,281				
			21	25	15,249				
			22	**32**	**15,321**				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	31	17,832	15	55	15,554	22	32	15,321	1
6	25	17,758	16	44	15,448	21	25	15,249	0
7	24	17,744	17	34	15,352	21, 30	25, 23	15,243.50	1
8	26	17,764	18	50	15,509	20, 21	27, 25	15,265	1
9	20	17,706	19	45	15,469	30	23	15,238	3
10	30	17,805	20	27	15,281	22	32	15,321	2
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40				

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: example

Job training program; selection into training depends only on age (Mixtape, Tables 5.8–5.9)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)			**Matched sample**			
Unit	Age	Earnings	Unit	Age	Earnings	Unit	Age	Earnings	Gap
1	23	17,728	11	49	15,499	30	23	15,238	0
2	26	17,770	12	54	15,566	20, 21	27, 25	15,265	1
3	23	17,736	13	47	15,487	30	23	15,238	0
4	27	17,777	14	49	15,501	20	27	15,281	0
5	31	17,832	15	55	15,554	22	32	15,321	1
6	25	17,758	16	44	15,448	21	25	15,249	0
7	24	17,744	17	34	15,352	21, 30	25, 23	15,243.50	1
8	26	17,764	18	50	15,509	20, 21	27, 25	15,265	1
9	20	17,706	19	45	15,469	30	23	15,238	3
10	30	17,805	20	27	15,281	22	32	15,321	2
			21	25	15,249				
			22	32	15,321				
			23	44	15,452				
			24	35	15,354				
			25	52	15,523				
			26	53	15,539				
			27	44	15,454				
			28	35	15,354				
			29	39	15,398				
			30	23	15,238				
Mean	25.5	17,762	Mean	41.8	15,427.40	Mean	26.1	15,265.95	0.9

Gap: $|X_i - X_{j(i)}|$, the matching discrepancy

Simple difference in outcomes: $17{,}762 - 15{,}427.40 = 2{,}334.60$

Nearest neighbor matching: $\hat{\delta}_{ToT} = 17{,}762 - 15{,}265.95 = 2{,}496.05$

Measuring "nearest"

With k covariates, we need a distance between X_i and X_j:

- **Euclidean**: $\|X_i - X_j\| = \sqrt{\sum_{n=1}^{k}(X_{ni} - X_{nj})^2}$
  - Depends on units: a 1-point gap in test score counts the same as a 1-year gap in birth year
- **Normalized Euclidean**: $\|X_i - X_j\| = \sqrt{\sum_{n=1}^{k} \frac{(X_{ni} - X_{nj})^2}{\hat{\sigma}_n^2}}$
  - Divides each covariate by its sample variance, so units don't matter
- **Mahalanobis**: $\|X_i - X_j\| = \sqrt{(X_i - X_j)' \hat{\Sigma}_X^{-1} (X_i - X_j)}$
  - $\hat{\Sigma}_X$: sample variance-covariance matrix of X
  - Also accounts for correlation between covariates

(Mixtape)

Nearest neighbors with two confounders

Now selection into training depends on age **and** high school GPA (Mixtape, Table 5.10)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)		
Unit	Age	GPA	Unit	Age	GPA
1	18	1.28	11	20	1.89
2	29	2.80	12	27	1.78
3	24	3.92	13	21	1.84
4	27	2.29	14	39	1.76
5	33	2.50	15	38	1.61
6	22	1.34	16	29	1.74
7	19	1.66	17	39	1.57
8	20	2.60	18	33	1.97
9	21	1.94	19	24	1.81
10	30	3.37	20	30	2.02
			21	33	1.64
			22	36	1.70
			23	22	1.66
			24	18	1.89
			25	43	1.45
			26	39	1.88
			27	19	1.86
			28	30	1.86
			29	51	1.96
			30	48	1.87
Mean	24.3	2.37	Mean	31.95	1.79

Nearest neighbors with two confounders

Now selection into training depends on age **and** high school GPA (Mixtape, Table 5.10)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)		
Unit	Age	GPA	Unit	Age	GPA
1	18	1.28	11	20	1.89
2	29	2.80	12	27	1.78
3	**24**	**3.92**	13	21	1.84
4	27	2.29	14	39	1.76
5	33	2.50	15	38	1.61
6	22	1.34	16	29	1.74
7	19	1.66	17	39	1.57
8	20	2.60	18	33	1.97
9	21	1.94	19	24	1.81
10	30	3.37	20	30	2.02
			21	33	1.64
			22	36	1.70
			23	22	1.66
			24	18	1.89
			25	43	1.45
			26	39	1.88
			27	19	1.86
			28	30	1.86
			29	51	1.96
			30	48	1.87
Mean	24.3	2.37	Mean	31.95	1.79

Who is the nearest neighbor of trainee 3 (age 24, GPA 3.92)?

Nearest neighbors with two confounders

Now selection into training depends on age **and** high school GPA (Mixtape, Table 5.10)

Trainees ($D_i = 1$)			**Comparison** ($D_i = 0$)		
Unit	Age	GPA	Unit	Age	GPA
1	18	1.28	11	20	1.89
2	29	2.80	12	27	1.78
3	**24**	**3.92**	13	21	1.84
4	27	2.29	14	39	1.76
5	33	2.50	15	38	1.61
6	22	1.34	16	29	1.74
7	19	1.66	17	39	1.57
8	20	2.60	**18**	**33**	**1.97**
9	21	1.94	**19**	**24**	**1.81**
10	30	3.37	**20**	**30**	**2.02**
			21	33	1.64
			22	36	1.70
			23	22	1.66
			24	18	1.89
			25	43	1.45
			26	39	1.88
			27	19	1.86
			28	30	1.86
			29	51	1.96
			30	48	1.87
Mean	24.3	2.37	Mean	31.95	1.79

Who is the nearest neighbor of trainee 3 (age 24, GPA 3.92)?

- Unit 19: same age, but GPA 2.11 points lower
- Unit 18: 9 years older, GPA 1.95 points lower
- Unit 20: 6 years older, GPA 1.90 points lower

"Nearest" depends on the distance metric

Trainee 3: age 24, GPA 3.92

Candidate	Age	GPA	Age gap	GPA gap	Euclidean	Normalized
19	24	1.81	0	2.11		
18	33	1.97	9	1.95		
20	30	2.02	6	1.90		

"Nearest" depends on the distance metric

Trainee 3: age 24, GPA 3.92

Candidate	Age	GPA	Age gap	GPA gap	Euclidean	Normalized
19	24	1.81	0	2.11	**2.11**	
18	33	1.97	9	1.95	9.21	
20	30	2.02	6	1.90	6.29	

Euclidean: $\sqrt{(\text{age gap})^2 + (\text{GPA gap})^2}$

- Unit 19 is nearest: a year of age counts the same as a GPA point

"Nearest" depends on the distance metric

Trainee 3: age 24, GPA 3.92

Candidate	Age	GPA	Age gap	GPA gap	Euclidean	Normalized
19	24	1.81	0	2.11	2.11	3.73
18	33	1.97	9	1.95	9.21	3.59
20	30	2.02	6	1.90	6.29	**3.42**

Euclidean: $\sqrt{(\text{age gap})^2 + (\text{GPA gap})^2}$

- Unit 19 is nearest: a year of age counts the same as a GPA point

Normalized Euclidean: $\sqrt{\frac{(\text{age gap})^2}{\hat{\sigma}^2_{age}} + \frac{(\text{GPA gap})^2}{\hat{\sigma}^2_{GPA}}}$, $\hat{\sigma}_{age} \approx 9.1, \hat{\sigma}_{GPA} \approx 0.57$

- Unit 20 is nearest: a 2.1-point GPA gap is almost 4 standard deviations

"Nearest" depends on the distance metric

Trainee 3: age 24, GPA 3.92

Candidate	Age	GPA	Age gap	GPA gap	Euclidean	Normalized
19	24	1.81	0	2.11	2.11	3.73
18	33	1.97	9	1.95	9.21	3.59
20	30	2.02	6	1.90	6.29	3.42

Euclidean: $\sqrt{(\text{age gap})^2 + (\text{GPA gap})^2}$

- Unit 19 is nearest: a year of age counts the same as a GPA point

Normalized Euclidean: $\sqrt{\frac{(\text{age gap})^2}{\hat{\sigma}^2_{age}} + \frac{(\text{GPA gap})^2}{\hat{\sigma}^2_{GPA}}}$, $\hat{\sigma}_{age} \approx 9.1, \hat{\sigma}_{GPA} \approx 0.57$

- Unit 20 is nearest: a 2.1-point GPA gap is almost 4 standard deviations

With more than one confounder, **the choice of metric determines the match**

Nearest neighbor matching with Euclidean distance

Match every trainee to its nearest neighbor using Euclidean distance on (age, GPA)

Trainees ($D_i = 1$)				**Nearest neighbor** $j(i)$					
Unit	Age	GPA	Earnings	Unit	Age	GPA	Earnings	Distance	$Y_i - Y_{j(i)}$
1	18	1.28	9,500	24	18	1.89	8,050	0.61	1,450
2	29	2.80	12,250	16	29	1.74	10,525	1.06	1,725
3	24	3.92	11,000	19	24	1.81	9,400	2.11	1,600
4	27	2.29	11,750	12	27	1.78	10,075	0.51	1,675
5	33	2.50	13,250	18	33	1.97	11,425	0.53	1,825
6	22	1.34	10,500	23	22	1.66	8,950	0.32	1,550
7	19	1.66	9,750	27	19	1.86	8,275	0.20	1,475
8	20	2.60	10,000	11	20	1.89	8,500	0.71	1,500
9	21	1.94	10,250	13	21	1.84	8,725	0.10	1,525
10	30	3.37	12,500	20	30	2.02	10,750	1.35	1,750
Mean	24.3	2.37		Mean	24.3	1.85			

- Every match has exactly the same age; GPA gaps remain (mean GPA 2.37 vs. 1.85)
- Euclidean distance is driven by age, the covariate with the largest units

Nearest neighbor matching with Euclidean distance

Match every trainee to its nearest neighbor using Euclidean distance on (age, GPA)

Trainees ($D_i = 1$)				**Nearest neighbor** $j(i)$					
Unit	Age	GPA	Earnings	Unit	Age	GPA	Earnings	Distance	$Y_i - Y_{j(i)}$
1	18	1.28	9,500	24	18	1.89	8,050	0.61	1,450
2	29	2.80	12,250	16	29	1.74	10,525	1.06	1,725
3	24	3.92	11,000	19	24	1.81	9,400	2.11	1,600
4	27	2.29	11,750	12	27	1.78	10,075	0.51	1,675
5	33	2.50	13,250	18	33	1.97	11,425	0.53	1,825
6	22	1.34	10,500	23	22	1.66	8,950	0.32	1,550
7	19	1.66	9,750	27	19	1.86	8,275	0.20	1,475
8	20	2.60	10,000	11	20	1.89	8,500	0.71	1,500
9	21	1.94	10,250	13	21	1.84	8,725	0.10	1,525
10	30	3.37	12,500	20	30	2.02	10,750	1.35	1,750
Mean	24.3	2.37	**11,075**	Mean	24.3	1.85	**9,467.50**		1,607.50

- Every match has exactly the same age; GPA gaps remain (mean GPA 2.37 vs. 1.85)
- Euclidean distance is driven by age, the covariate with the largest units

Nearest neighbor matching: $\hat{\delta}_{ToT} = 11{,}075 - 9{,}467.50 = 1{,}607.50$

Nearest neighbor matching with normalized Euclidean distance

Match every trainee using normalized Euclidean distance on (age, GPA)

Trainees ($D_i = 1$)				**Nearest neighbor** $j(i)$					
Unit	Age	GPA	Earnings	Unit	Age	GPA	Earnings	Distance	$Y_i - Y_{j(i)}$
1	18	1.28	9,500	23	22	1.66	8,950	0.80	550
2	29	2.80	12,250	20	30	2.02	10,750	1.38	1,500
3	24	3.92	11,000	20	30	2.02	10,750	3.42	250
4	27	2.29	11,750	20	30	2.02	10,750	0.58	1,000
5	33	2.50	13,250	20	30	2.02	10,750	0.91	2,500
6	22	1.34	10,500	23	22	1.66	8,950	0.57	1,550
7	19	1.66	9,750	23	22	1.66	8,950	0.33	800
8	20	2.60	10,000	11	20	1.89	8,500	1.26	1,500
9	21	1.94	10,250	11	20	1.89	8,500	0.14	1,750
10	30	3.37	12,500	20	30	2.02	10,750	2.39	1,750
Mean	24.3	2.37		Mean	25.6	1.89			

- GPA gaps shrink (mean GPA 2.37 vs. 1.89), but ages now differ (24.3 vs. 25.6)
- Unit 20 is the match for 5 trainees: we match **with replacement**

Nearest neighbor matching with normalized Euclidean distance

Match every trainee using normalized Euclidean distance on (age, GPA)

Trainees ($D_i = 1$)				**Nearest neighbor** $j(i)$					
Unit	Age	GPA	Earnings	Unit	Age	GPA	Earnings	Distance	$Y_i - Y_{j(i)}$
1	18	1.28	9,500	23	22	1.66	8,950	0.80	550
2	29	2.80	12,250	20	30	2.02	10,750	1.38	1,500
3	24	3.92	11,000	20	30	2.02	10,750	3.42	250
4	27	2.29	11,750	20	30	2.02	10,750	0.58	1,000
5	33	2.50	13,250	20	30	2.02	10,750	0.91	2,500
6	22	1.34	10,500	23	22	1.66	8,950	0.57	1,550
7	19	1.66	9,750	23	22	1.66	8,950	0.33	800
8	20	2.60	10,000	11	20	1.89	8,500	1.26	1,500
9	21	1.94	10,250	11	20	1.89	8,500	0.14	1,750
10	30	3.37	12,500	20	30	2.02	10,750	2.39	1,750
Mean	24.3	2.37	**11,075**	Mean	25.6	1.89	**9,760**		1,315

- GPA gaps shrink (mean GPA 2.37 vs. 1.89), but ages now differ (24.3 vs. 25.6)
- Unit 20 is the match for 5 trainees: we match **with replacement**

Nearest neighbor matching: $\hat{\delta}_{ToT} = 11{,}075 - 9{,}760 = 1{,}315$ (Euclidean: 1,607.50)

Matching

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Important extensions

Logistic regression

Review: matching

Variables:

- Y_i: Outcome variable
- D_i: Binary treatment dummy variable
- X_i: Covariates

Under conditional independence assumption (CIA) $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$, we can construct causal matching estimators:

$$\begin{aligned}\delta_{ToT} &= \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x|D_i = 1) \\ \delta_{ATE} &= \mathbf{E}[Y_i(1) - Y_i(0)] \\ &= \sum_x \delta_x \mathbf{P}(X_i = x)\end{aligned}$$

where:

$$\delta_X = \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0]$$

Note: matching and regression are both weighted averages of covariate specific differences

- Problem: with many covariates, matching on X_i runs into the curse of dimensionality
- Propensity score: match on a **single** number instead of the whole vector X_i

Motivating example

"Growth mindset" example (heavily borrows from: link)

- If you think your intelligence is fixed, there is no incentive to study; academic outcomes are worse
- If you have a "growth mindset," there are returns to intelligence; your academic outcomes are better

The National Study of Learning Mindsets

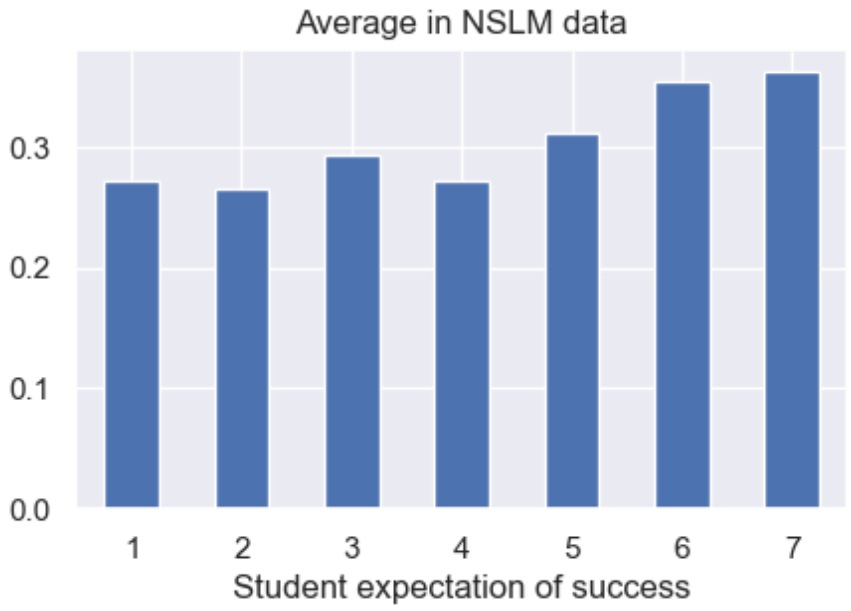
- Evaluate the impact of a nudge-like intervention designed to instill students with a growth mindset on student achievement
- Students randomly given the opportunity to participate in "growth mindset" seminar in high school; academic outcomes in college were then recorded
- Problem: students with the highest expectation of success were more likely to receive the treatment
  - Opportunity to participate in the seminar was random
  - Those who moved forward with the opportunity may have been more ambitious than those who did not

Athey and Wager (2019) generate simulated data: link

Motivating example (cont)

Variables:

- Y_i: Outcome variable (achievement score in college)
- D_i: Binary treatment dummy variable (participating in "growth mindset" seminar)
- X_i: Covariates (individual and school demographics; family characteristics; survey about expectations of future success)



More ambitious students may be more willing to go to the seminar, leading to selection bias. How can we correct for that?

Propensity Score Theorem

Define the propensity score (or balancing score) to be:

$$\begin{aligned} p(X_i) =& \mathbf{E}[D_i|X_i] \\ =& \sum_{d=0}^{1} d \cdot \mathbf{P}(D_i = d|X_i) \\ =& 0 \cdot \mathbf{P}(D_i = 0|X_i) + 1 \cdot \mathbf{P}(D_i = 1|X_i) \\ =& \mathbf{P}(D_i = 1|X_i) \end{aligned}$$

Note: the propensity score equals the probability of treatment for binary treatments

Suppose the CIA holds such that $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$. Then $\{Y_i(0), Y_i(1)\} \perp D_i|p(X_i)$ (Proof: Rosenbaum and Rubin 1983; MHE Theorem 3.3.1)

What does this mean?

- You don't need to directly control for confounders X to achieve CIA
- Instead, it is sufficient to control for a single variable, $p(X_i)$

Balancing property: Given propensity score:

$$p(X_i) = \mathbf{E}[D_i|X_i] = \mathbf{P}(D_i = 1|X_i)$$

Treatment is independent of covariates conditional on the propensity score:
$D_i \perp X_i|p(X_i)$

Proof:
Because D_i is binary, it is enough to show that:

$$\mathbf{E}[D_i|X_i, p(X_i)] = \mathbf{E}[D_i|p(X_i)]$$

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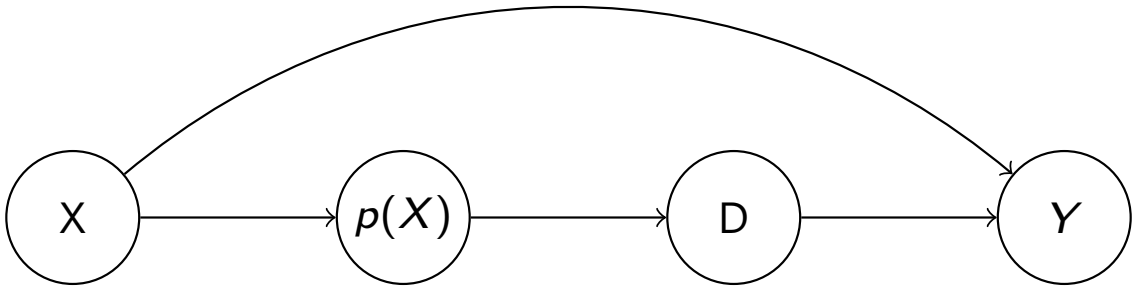
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Because $p(X_i) = p(X_i)$, the statement is true.



- If I know what $p(X)$ is, then X alone doesn't tell me anything else about what the probability of treatment is
- Controlling for $p(X)$ acts the same way as controlling for X directly
- Unlike the CIA, the balancing property is **testable**: within narrow bands of $p(X)$, treated and untreated units should have similar covariate means

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Propensity scores and treatment effects

How are propensity scores used in practice?

- First, estimate the propensity score
  - Use a parametric model, like a logit or probit model
- Next generate the effect of treatment
  - Match on estimated score (direct propensity score matching)
  - OR use weighting scheme described next

Direct propensity score matching

Given a propensity score, $p(X_i)$, and the CIA $\{Y_i(0), Y_i(1)\} \perp D_i|X_i$:

$$\begin{aligned}\delta_{ToT} =& \mathbf{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ =& \mathbf{E}\left[\mathbf{E}[Y_i|p(X_i), D_i = 1] - \mathbf{E}[Y_i|p(X_i), D_i = 0]|\, D_i = 1\right]\end{aligned}$$

Just like we did for matching, you can

- Stratify the sample into values of $p(X_i)$
- Find treatment effects conditional on similar $p(X_i)$
- Weight these treatment effects by the distribution of $p(X_i)$ conditional on treatment

In practice: nearest neighbor matching on the estimated score $\hat{p}(X_i)$

- $\hat{p}(X_i)$ is a single number, so there is no distance metric to choose

Propensity score weights

Alternative: estimate treatment effects using propensity score weights. Final equation:

$$\mathbf{E}[Y_i(1) - Y_i(0)] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1-p(X_i))}\right]$$

We'll go through the math. Motivation for this method:

- Avoids the matching step
- The concept of inverse probability weighting is important in more advanced topics

To prove this, we'll have to prove a few interim steps

Fact #1:

$$\mathbf{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbf{E}[Y_i(1)]$$

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Next, recall $Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i$:

$$Y_i D_i = (Y_i(0) + (Y_i(1) - Y_i(0))D_i)D_i$$

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Using the above equality, the conditional independence assumption, and the definition of the propensity score:

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\end{aligned}$$

Fact #2:

$$\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_i(0)]$$

Proof: Recall $Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i$, and note $D_i(1-D_i) = 0$ for $D_i = 0$ and $D_i = 1$:

$$Y_i(1-D_i)$$

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$$Y_i(1-D_i) = (Y_i(0) + (Y_i(1) - Y_i(0))D_i)(1-D_i)$$

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Using the above equality, the conditional independence assumption, and the definition of the propensity score:

$$\mathbf{E}[Y_i(1-D_i)|X_i] = \mathbf{E}\left[\, Y_i(0)(1-D_i)|\, X_i\right]$$

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\end{aligned}$$

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$$\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_i(0)]$$

Proof (cont.):

We've seen:

$$\mathbf{E}[Y_i(1-D_i)|X_i] = \mathbf{E}[Y_i(0)|X_i](1-p(X_i))$$

Now we can estimate $\mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right]$:

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We've seen

$$\mathbf{E}\left[\frac{Y_i D_i}{p(X_i)}\right] = \mathbf{E}[Y_i(1)], \qquad \mathbf{E}\left[\frac{Y_i(1-D_i)}{1-p(X_i)}\right] = \mathbf{E}[Y_i(0)]$$

Average treatment effects can be estimated from:

$$\begin{aligned}
\mathbf{E}[Y_i(1) - Y_i(0)] =& \mathbf{E}\left[\frac{Y_i D_i}{p(X_i)} - \frac{Y_i(1-D_i)}{1-p(X_i)}\right] \\
=& \mathbf{E}\left[\frac{Y_i D_i(1-p(X_i)) - Y_i(1-D_i)p(X_i)}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i(1-p(X_i)) - (1-D_i)p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - D_i p(X_i) - p(X_i) + D_i p(X_i)\right]}{p(X_i)(1-p(X_i))}\right] \\
=& \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1-p(X_i))}\right]
\end{aligned}$$

Horvitz-Thompson Propensity score weighting:

$$\mathbf{E}[Y_i(1) - Y_i(0)] = \mathbf{E}\left[\frac{Y_i\left[D_i - p(X_i)\right]}{p(X_i)(1-p(X_i))}\right]$$

- Uses propensity score weighting to recover average treatment effects
- Using the propensity score rather than identification by conditioning on X is a useful empirical strategy when X is high-dimensional and $p(X)$ is available (or can be approximated accurately)
- MHE shows how this can be considered similar to the regression estimator

Matching

Matching estimators

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Inverse Probability of Treatment Weighting

We've seen:

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Consider the first element:

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- $Y_i D_i$: Isolates Y_i that are treated

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Consider the first element:

- $Y_i D_i$: Isolates Y_i that are treated
- $1/p(X_i)$: Inverse probability of being treated
  - Probabilities are between 0 and 1
  - Low probability events $\implies$ inverse probability high

$$p(X) = 0.1 = \frac{1}{10} \implies \frac{1}{p(X)} = \frac{1}{1/10} = 10$$

  - High probability events $\implies$ inverse probability low

$$p(X) = 0.8 = \frac{4}{5} \implies \frac{1}{p(X)} = \frac{1}{4/5} = 1.25$$

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The same reasoning applies to the second element

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The same reasoning applies to the second element

IPTW: valuable in a clinical setting to generate balanced data

Structural model and connection to machine learning

Note: the following is from *Business Data Science* by Matt Taddy, Chapter 6

Consider a common structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Note:

- $\epsilon|D, X = 0$ and $v|X = 0$ encode CIA

Two-step **linear treatment effects (LTE)** model

- First estimate treatment D_i
- Then estimate treatment effects γ

Dimensionality starts to matter here

- If n (number of observations) are much greater than p (number of X variables), you can use regression models
- If you don't have $n >> p$, traditional methods break down

Estimation of LTE

Structural model:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D = X'\tau + v, \quad v|X = 0$$

Estimation procedure:

1. Estimate $\mathbf{E}[D|X] = X'\tau$. Collect fitted values $\hat{D} = X'\hat{\tau}$
2. Estimate:
$$\mathbf{E}[Y|X, D] = \hat{D}\psi + D\gamma + X'\beta$$

The ψ is the treatment effect of D on Y.

LTE and propensity score matching

Structural model in propensity score models:

$$Y = D\gamma + X'\beta + \epsilon, \quad \epsilon|D, X = 0$$
$$D \sim \text{Bernoulli}\left(p(X)\right)$$
$$p(X) = \frac{e^{X'\tau}}{1 + e^{X'\tau}}$$

- D is modeled from a Bernoulli distribution (i.e. coin toss) with probability p
- The probability that $D = 1$ is modeled using a **logistic regression**

Once you estimate the propensity score $p(X)$, you can find treatment effects by:

1. Fitting the linear regression: $\mathbf{E}[Y|D, \hat{p}(X)] = \alpha + D\gamma + \hat{p}(X)\psi$
2. Constructing propensity score weighting estimates: $\mathbf{E}\left[\left.\frac{Y}{p(\hat{X})}\right| D\right] = \alpha + D\gamma$
3. Matching

Hahn (1998) shows there is a cost to propensity score matching in terms of asymptotic efficiency

- What is the advantage of, say, controlling for propensity score instead of controlling for covariates in a regression?

MHE : "An essential element of propensity score methods is the use of prior knowledge for dimension reduction. The statistical payoff is an improvement in finite sample behavior. If you're not prepared to smooth, restrict, or otherwise reduce the dimensionality of the matching problem in a manner that has real empirical consequences, then you might as well go for full covariate matching or saturated regression control."

- This is very prescient! Now, in the era of big data, dimensionality reduction is key
- Propensity scores have a new value in double-robust machine learning; see chapter 9 of `https://chapters.causalml-book.org/CausalML_chap_9.pdf`

Matching

Matching estimators

Control for covariates using the Propensity Score

Propensity scores and treatment effects

Important extensions

Logistic regression

Why logistic regression

- We have to use some estimation procedure to predict propensity scores $p(X)$
- Logistic regression is a common method
- Quick review of logistic regression and generalized linear models

Linear regression

Assume we have data $\{y_i, x_{1i}, x_{2i}, \ldots, x_{ki}\} = \{y_i, \mathbf{x}_i\}$. A linear regression model takes the form:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \epsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n$$

We can stack these equations for all n equations in the data:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

To estimate this model using least squares, find the coefficient matrix $\boldsymbol{\beta}$ that minimizes the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} x_{ji}\beta_j \right)^2$$

A more math-y way of saying this using norm notation:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} ||\mathbf{y} - \mathbf{X}\boldsymbol{\beta}||^2$$

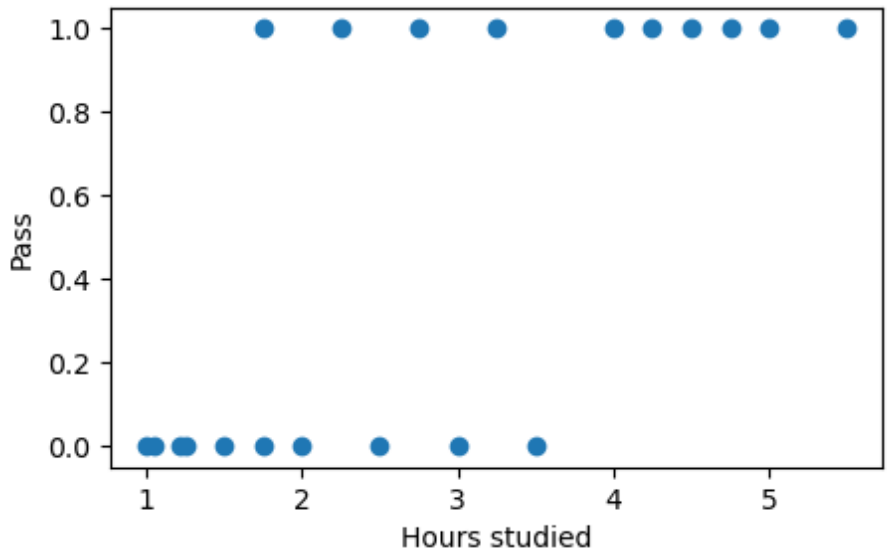
- Just think about the norm notation $||\cdot||$ communicating size in a multidimensional space

Limitations of linear model

We are trying to predict treatment, D_i

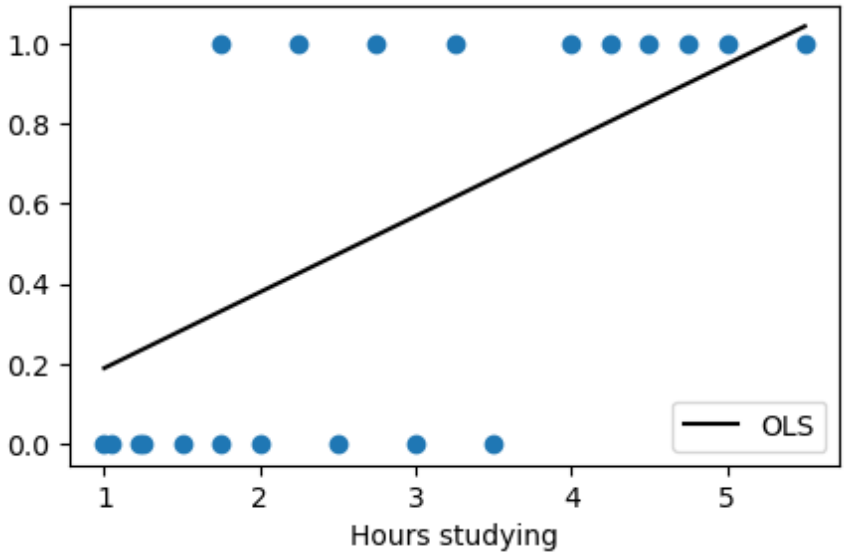
- Always takes on values equal to zero or 1
- Do we really want a linear model?

Consider data on hours studied and passing/failing a test ($D = 1$ implies a student passed)



OLS

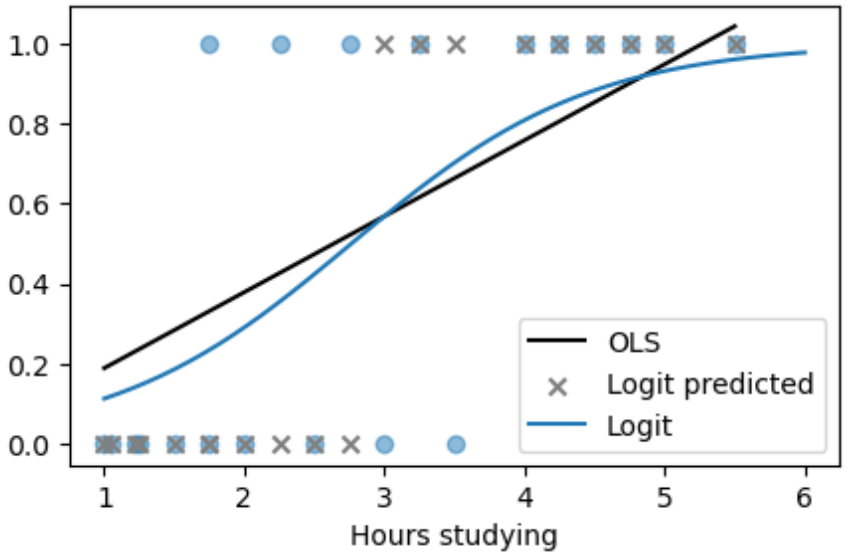
We could estimate using a linear model:



- Line isn't a great fit

Logit

We could estimate using a logistic model:



- Logistic line is a better fit
- Less important: fitted values are probabilities between 0 and 1